



— PROBLEM STATEMENT DOCUMENT

AI-Driven Cyclone & Coastal Disaster Early Warning System

An Agentic AI-inspired coastal disaster intelligence platform delivering hyperlocal cyclone tracking, fishermen safety alerts, coordinated evacuation and relief planning, and post-disaster damage assessment for Gujarat's coastline.

🌐 Domain: Disaster Management

🤖 5 Agentic AI Agents

🎓 IBM University Engagement

1,600+

KM COASTLINE

12

COASTAL DISTRICTS

05

AI AGENTS

07

RESPONSE STAGES

1 Project Overview

CoastGuard AI is a coastal disaster intelligence platform concept built around an Agentic AI architecture. It is designed to connect cyclone prediction, fishermen safety, evacuation planning, relief coordination, and post-disaster assessment into a single, coordinated decision-support layer for Gujarat's coastal districts.

2 The Challenge

Gujarat's 1,600+ km coastline is highly vulnerable to cyclones such as Tauktae and Biparjoy. Coastal communities and fishermen often face delayed or generalized warnings, which can lead to preventable loss of life, boats, property, and infrastructure. Traditional disaster-response processes can suffer from:



Delayed Warnings

Slow chain between detection and ground-level alerts.



Hyperlocal Gaps

Lack of village and coastal micro-risk intelligence.



Vessel Monitoring

Difficulty tracking fishermen and vessels at sea.



Evacuation Gaps

Uncoordinated evacuation planning and routing.



Resource Mismatch

Relief resources not matched to local requirements.



Slow Assessment

Delayed post-disaster damage assessment.

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How can Agentic AI be used to transform cyclone and coastal disaster management by providing hyperlocal cyclone intelligence, timely fishermen safety alerts, coordinated evacuation and relief planning, and faster post-disaster damage assessment for Gujarat's coastal districts?



Hyperlocal Coastal Risk Radar

CoastGuard AI's command-center concept continuously overlays live cyclone position, wind intensity, and district-level risk zones — turning fragmented meteorological data into a single, hyperlocal picture of coastal risk that district officials, disaster-response teams, and fishermen can act on quickly.

Objective & Proposed Solution

3 Objective

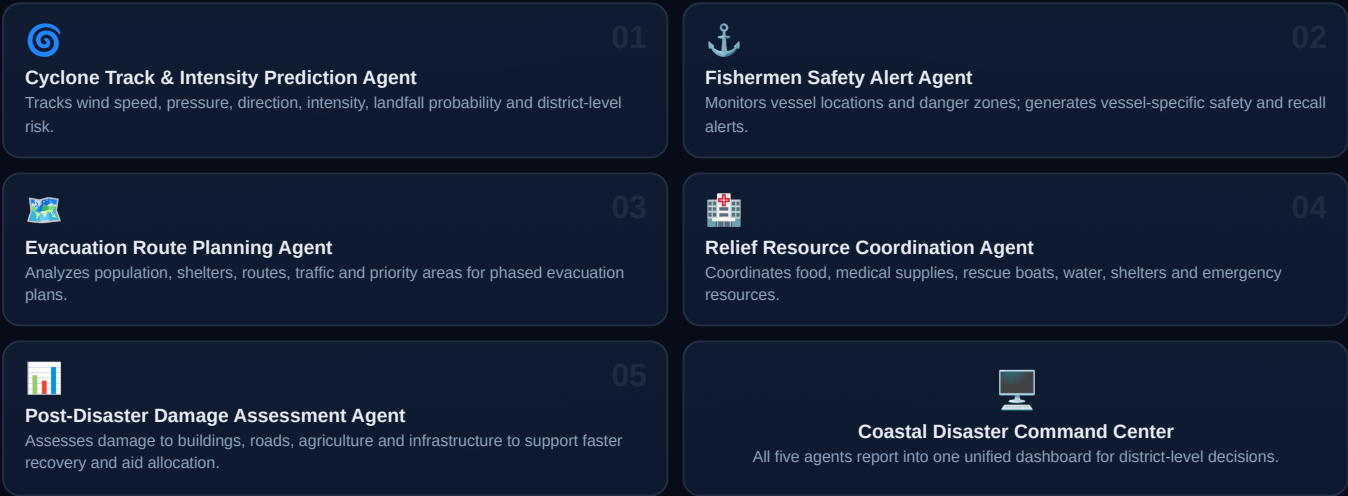
Build an **Agentic AI solution** that provides hyperlocal cyclone tracking, fishermen safety alerts, coordinated evacuation planning, relief resource coordination, and post-disaster damage assessment for coastal districts of Gujarat.

4 Proposed Solution — Disaster Response Workflow

CoastGuard AI connects the complete disaster-response cycle through a unified agentic intelligence layer:



5 Agentic AI Architecture — Five Specialized Agents

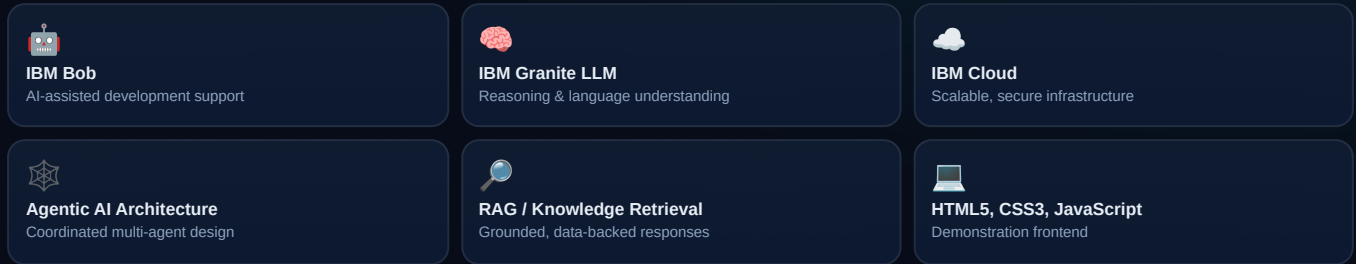


How the Agents Connect



Technology Stack & Expected Impact

6 Technology Stack



7 Expected Impact

- ✓ Faster identification of cyclone risks
- ✓ Improved fishermen safety at sea
- ✓ More efficient emergency resource allocation
- ✓ Unified disaster-management decision support across the full response cycle
- ✓ More targeted warnings for vulnerable communities
- ✓ Better-coordinated evacuation planning
- ✓ Faster post-disaster damage assessment

8 Innovation & Uniqueness

The uniqueness of CoastGuard AI lies in its **multi-agent architecture**. Instead of relying on a single AI system for every disaster-management task, five specialized AI agents collaborate across prediction, fishermen safety, evacuation, relief and recovery — connecting the entire disaster-response cycle through one coordinated intelligence layer.

9 Future Scope

Satellite & Drone Integration

Live satellite feeds and post-cyclone drone imagery for sharper landfall prediction and faster damage assessment.

Multilingual Voice & SMS Alerts

Regional-language voice and SMS alerts for fishermen and communities with limited connectivity.



Prototype Disclaimer: CoastGuard AI is a frontend demonstration / hackathon prototype. All alerts, risk scores, maps, vessel positions, cyclone information and AI analyses shown on the demo website are entirely simulated and fictional — this is NOT a real-time emergency warning system. In an actual emergency, always follow official advisories from IMD, NDMA and NDRF.

One Intelligence Layer. Complete Disaster Response.

From the first cyclone alert to the last relief truck, CoastGuard AI's five agents are designed to work as one coordinated system — helping Gujarat's coastal districts move from reactive response to proactive, hyperlocal disaster preparedness.

IBM University Engagement

Program Submission

Agentic AI + Langflow

Built with IBM Granite

Disaster Management

Domain Focus